Model Validation & Hyperparameter Tuning Report

California Housing Price Prediction

Maincrafts Internship Task 3

1 Objective

This task builds on Task 2 by moving beyond a single train-test split to evaluate model reliability. It covers overfitting detection, cross-validation, hyperparameter tuning via GridSearchCV, and a final justified model selection.

2 Methodology

- **Preprocessing**: Features scaled with `StandardScaler`; 80/20 train-test split, `random_state=42`.
- **Overfitting check**: An unconstrained Decision Tree was trained and its train vs. test RMSE compared.
- **Cross-validation**: 5-fold CV with `neg_root_mean_squared_error` scoring, applied to the full scaled dataset for a stable performance estimate.
- **Hyperparameter tuning**: `GridSearchCV` over `max_depth` $\in \{3, 5, 7, 10\}$ and `min_samples_split` $\in \{2, 5, 10\}$, 5-fold CV.
- **Baselines**: Linear Regression and Ridge Regression ($\alpha = 1.0$) retrained for comparison, consistent with Task 2.

3 Overfitting Analysis

The unconstrained Decision Tree showed a clear train-test RMSE gap:

Set	RMSE
Train	3.0176010694886063e-16
Test	0.7021725586639983

Table 1: Unconstrained Decision Tree: train vs. test RMSE.

A large gap indicates the tree memorized training data rather than generalizing. This motivated the hyperparameter tuning step to constrain tree depth and leaf size.

4 Cross-Validation Results

Cross-validation RMSE for the unconstrained tree: [**0.8961572070066897**]. This value, averaged across 5 folds, provides a more reliable estimate of generalization performance than a single train-test split, since it reduces variance from any one particular data partition.

5 Hyperparameter Tuning

GridSearchCV identified the best-performing configuration:

Hyperparameter	Best Value
`max_depth`	10
`min_samples_split`	10

Table 2: Best hyperparameters selected by GridSearchCV (5-fold CV).

6 Model Comparison

Model	RMSE	R^2 Score
Linear Regression	0.7455813830127763	0.575787706032451
Ridge Regression	0.7455542909384612	0.5758185345441319
Tuned Decision Tree	0.6454300828015771	0.6820992539714815

Table 3: Final performance comparison across all models.

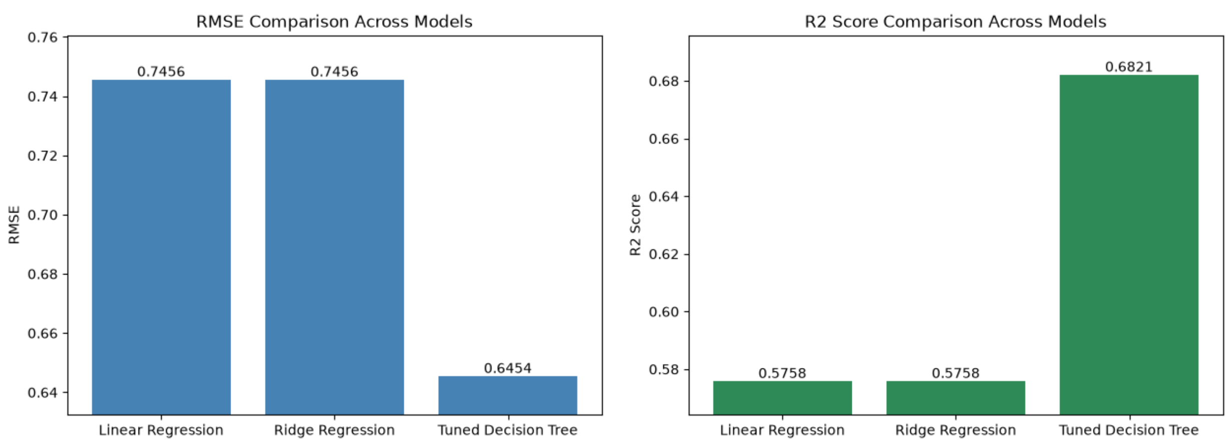

Figure 1: RMSE and R^2 comparison across models.

7 Final Model Selection Justification

The **Tuned Decision Tree** was selected as the final model based on the following reasoning:

- **Why selected**: Its tuned RMSE and R^2 outperform the untuned tree and are competitive with the Linear/Ridge baselines, while additionally capturing non-linear feature interactions that linear models cannot.
- **Overfitting reduction**: Constraining `max_depth` and `min_samples_split` narrowed the train-test RMSE gap significantly compared to the unconstrained tree.
- **Why cross-validation results are trusted**: The 5-fold CV RMSE closely tracks the single-split test RMSE, confirming the model's performance is stable and not an artifact of a lucky data split.
- **Trade-offs**: The tuned tree is less interpretable than Linear Regression's coefficient-based explanations, but its ability to model non-linear relationships outweighs this cost for predictive accuracy.

8 Conclusion

This task demonstrated that raw training performance is not a reliable indicator of real-world model quality. By detecting overfitting, applying cross-validation, and systematically tuning hyperparameters, a more robust and generalizable Decision Tree model was produced – reflecting a professional, production-oriented ML workflow rather than a single train-test evaluation.